

Homie Rektifier: Mechanical Rectification

EID-103 Principles of Design

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Introduction

The engineers set out to design a mechanical rectifier capable of converting non-constant unidirectional rotary motion to constant velocity unidirectional rotary motion. The rectifier was meant to serve as a physical counterpart to an electrical rectifier that can convert alternating current (AC) to direct current (DC). The primary challenge was to create a rectifying system that can convert mechanical input without any motors or electronics. In addition, the system was designed with strict performance, safety, and integration requirements. Specifically, the rectifier must connect to the input motor shaft by engaging with an encoder gear and be capable of measuring the input and output speeds. To ensure ease of deployment, the system must be able to assemble and disassemble onto the aluminum breadboard in under 5 minutes. Additionally, it was required to reach steady-state operation in less than ten seconds and perform reliably across three distinct input frequencies (0.25 Hz, 0.5 Hz, and 1.0 Hz).

Brainstorming Solutions

The design challenge emphasized mechanical innovation and precise construction. Developing a mechanical rectifier that relied solely on passive mechanical components, rather than electrical motors, required a creative and efficient design. Additionally, keeping the entire system confined to the 8" x 8" aluminum breadboard required the design to be compact and precise. Thus, our group approached this task by first coming up with potential solutions to the task.

The first design, “Homie Rektifier”, featured a central sliding gear that, depending on the direction of rotation, would slide in place to connect the input gears with the output gear. To alternate the direction that the sliding gear spins, a spring-loaded lever was integrated into the system. This lever is hit by a cam connected to the encoder gear. As the cam rotates, it presses the lever down to spin the first gear clockwise. When the lever is released, the lever spins the gear in the opposite direction. This cyclical motion of the lever engages the sliding gear so it can only make contact with the final gear when spinning in the clockwise direction. The final gear then engages with an output shaft equipped with a flywheel, rectifying the inconsistent input rotational velocity into continuous output rotational velocity.

The second design, “L'Hopital's Limiter”, was a massive gear system that would be connected from the input gear directly to the output gear that would rotate the rod. The idea behind this design was the gear ratio would be so large that the varying motion from the input motor would be mitigated by all the gears, resulting in the rod moving with such small velocity it would essentially be constant.

The third design, “Derivative Drive”, would have used a magnetic coupler to attenuate the effects of the variation in angular velocity. Because the two magnets would rotate together at lower velocities, but disconnect at higher velocities, it would be possible to eliminate most of the higher, resulting in more constant rotation. A flywheel would be used in tandem with the coupler to provide additional rotational inertia.

	Homie Rektifier (sliding gear)	L'Hopital's Limiter	Derivative drive
Size	7	1	5
Time to assemble onto plate	8	9	5

Response time	8	2	7
Efficiency	9	2	3
Total	32	14	20

Using a decision matrix based on criteria such as size, assembly time, and rectification efficiency, the sliding gear mechanism emerged as the most optimal option. The design was substantially more compact and was projected to rectify better than any alternative design.

Discussion of the chosen solution

A major principle when brainstorming and designing any of our initial prototypes was ensuring power input was never directly connected to the output shaft mechanically. In other words, the input and output could not be connected merely by gears, as a translation of power from gear to gear directly can at most only adjust the velocity of the output shaft and not frequency. As seen in the above section with the design "L'Hopital's Limiter", an attempt was made to utilize a direct connection; however, the design was regarded as inefficient and thus discarded. Therefore, we decided to innovate a method in which input and output rotation was never directly connected.

With this principle in mind, we brainstormed methods in which input power can indirectly be translated to an output shaft. A thought we devised was designing a machine that translates constant rotational motion into pulses of power. An example we thought of when conceiving this concept is as follows:

For example, when attempting to replace a tire on their bicycle, one must first flip the bike so it rests with its seat and handlebars on the ground. In this position, the bike wheels are

free to spin. One may notice that applying a force with their hand to a wheel of the bike results in that wheel spinning, even after they have stopped applying any sort of force. This is due to the high rotational inertia of a bicycle wheel. Additionally, any subsequent applications of force on the wheel will continually increase its rotational velocity; however, after a certain point, the force being applied onto the spinning wheel is insufficient enough to speed up the wheel. Therefore, regardless of any additional applications of power of the same or similar magnitude, the wheel will remain spinning at a constant velocity.

With this example in mind, we derived a system which translates non-constant input rotation to constant output rotation where the primary form of translating constant rotational power into pulses was done through a sliding gear concept.

Design Process

Sliding Gear Subsystem:

The design process began with a visual prototype of the subsystem around the sliding gear to establish an initial understanding of its structure. By utilizing paper, wooden skewers, rubber bands, and KNEX gears, the following visual prototype of the sliding gear subsystem was created.

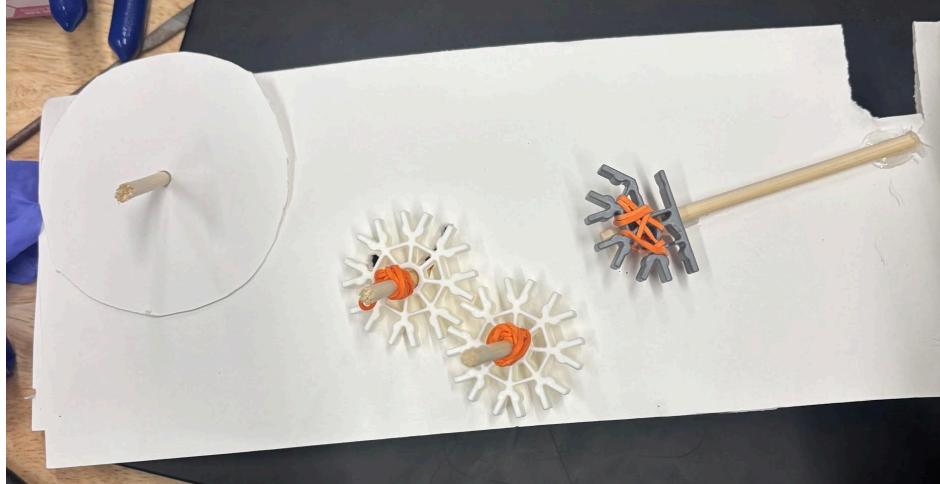


Figure 1: First sliding gear subsystem prototype

After completing the visual prototype, initial CAD models were designed and assembled in Onshape. A physical prototype of the sliding gear system was then 3D printed using Prusa Mini 3D printers as a proof of concept to demonstrate the feasibility of the mechanism.



Figure 2: Second sliding gear subsystem prototype

Upon seeing the sliding gear mechanism function as envisioned, a first prototype of the gear system with a stand that can be mounted to the breadboard was created.

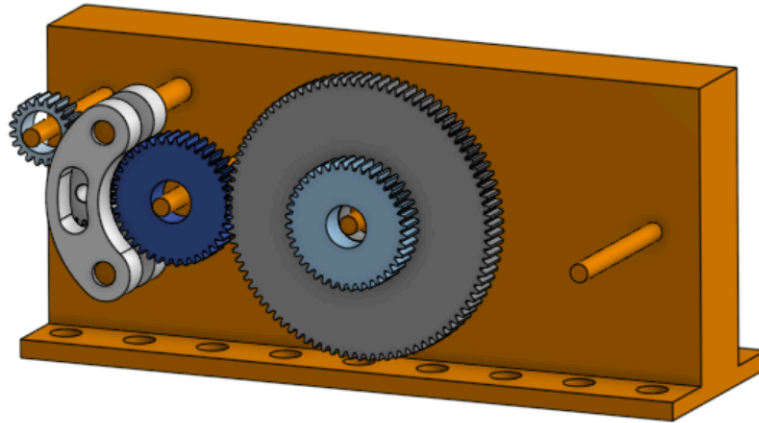


Figure 3: Third sliding gear subsystem prototype

The first prototype, as shown in figure 3, functioned better than expected, with the gears meshing properly and the sliding gear moving in its intended direction. However, the gears, specifically the final gear and the sliding gear, were extremely small, which made the output rotation weaker than intended. Additionally, the use of a single support wall resulted in the rods bending and the gears slipping along the rods.

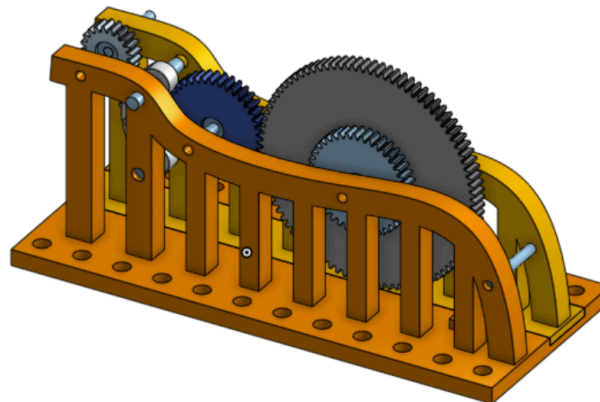


Figure 4: Fourth sliding gear subsystem prototype

In the second prototype, the gear sizes were increased by 150%, and the stand includes a vertical support bracket that can slide into place. This design effectively prevented the rods from

bending, but the larger gears meant that the gear system was too long to completely fit on the breadboard.

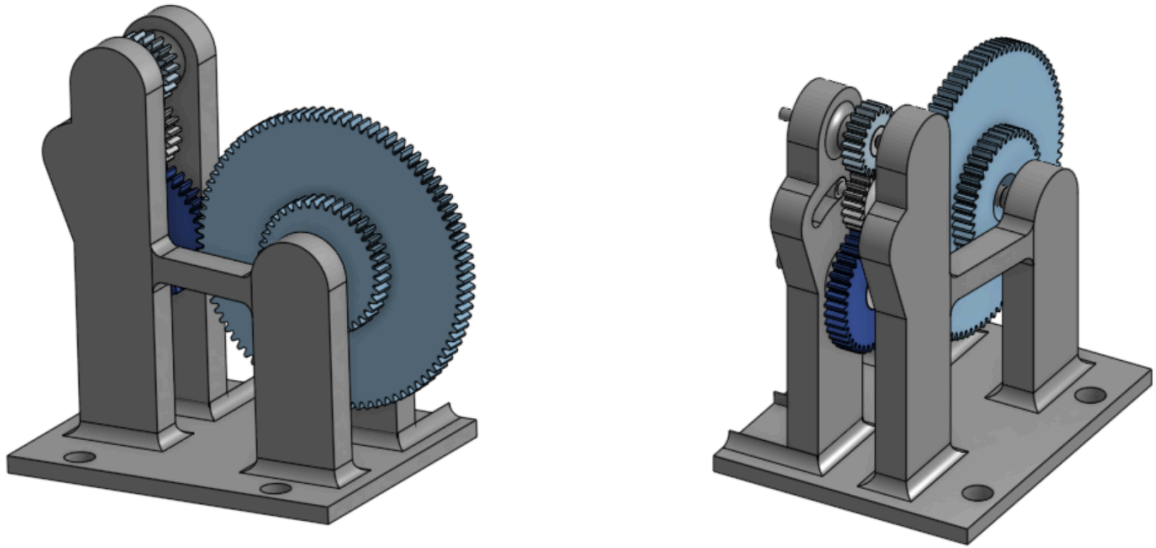


Figure 5: Final sliding gear subsystem

The final design addressed previous issues by repositioning the sliding gear and the final gear above the second input gear. This new arrangement allowed the sliding gear to move horizontally rather than vertically, reducing the influence of gravity on its motion. As a result, the gear slid with less resistance, improving responsiveness during rectification.

Cam Mechanism and Lever Gear:

Having the sliding gear concept completely developed, the next challenge was to translate power from the motor shaft into the sliding gear subsystem. Translating the rotational power from the input shaft to the sliding gear section was also a challenge in itself, as the

breadboard provided limited space. Therefore, the second major subsystem of Homie Rektifier consists of a cam mechanism and lever gear with its main purpose serving as a means of connecting the sliding gear subsystem to the input power.

The sliding gear subsystem previously discussed was designed to take varying inputs. Essentially, the output gear of the sliding gear subsystem would rotate from an input of the first major gear repeatedly spinning in opposite directions. In order to direct power from the input motor to the sliding gear subsystem, this subsystem also needed to translate the rotational motion into "up and down" motion. The system we devised to implement this concept consisted of a lever pivoting around its center with a subsection of a gear on one end. The "lever gear" then meshes with the smaller gear of the large input gear of the sliding gear subsystem. In order to activate the lever from a rotational input, a transfer of rotational motion to reciprocatory motion via a cam was implemented.

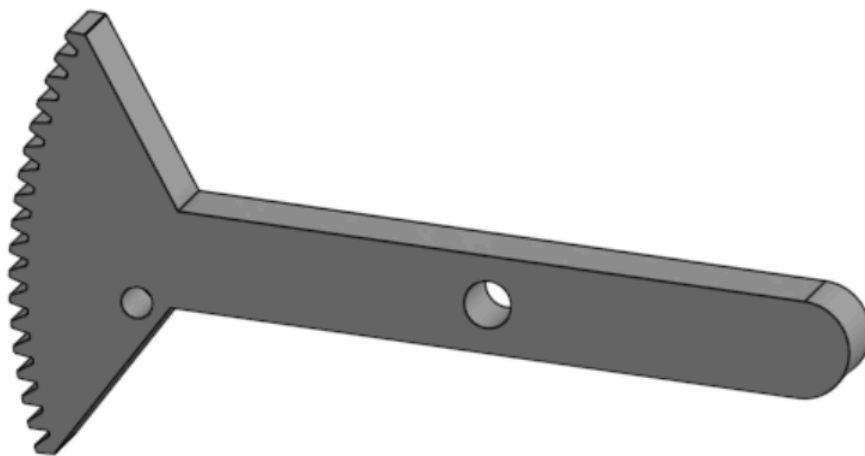


Figure 6: Lever Gear first iteration

The first iteration of the lever gear was designed using crude geometry, that being a simple rectangular lever with a fraction of a gear attached to the end. Two additional holes were removed from the lever in order to fit an axle which served as a pivot and a hole for an elastic material, such as a spring or rubber band, to be passed through. This design was evaluated as faulty after manufacturing and assembling early prototypes, as the opposite end lacked an effective surface for a cam to strike it.

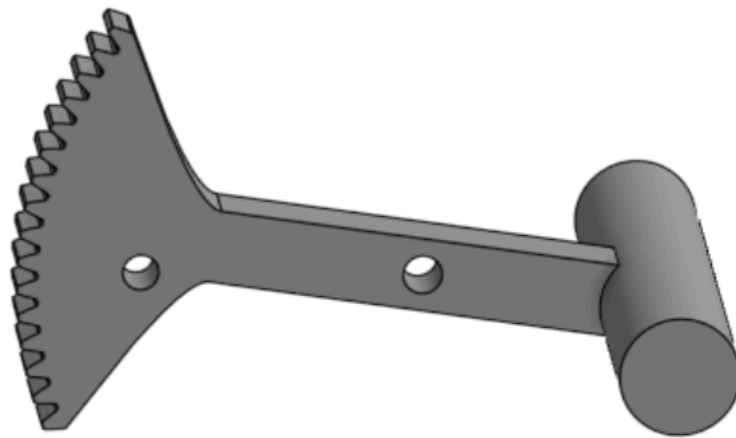


Figure 7: Lever Gear second iteration

After evaluating significant flaws with the first iteration, the second iteration incorporated a larger surface for a cam to strike it and thus provide power. However, the spring hole was still too small and ill positioned and needed improvements; furthermore, the small pivot hole imposed significant friction on the lever.

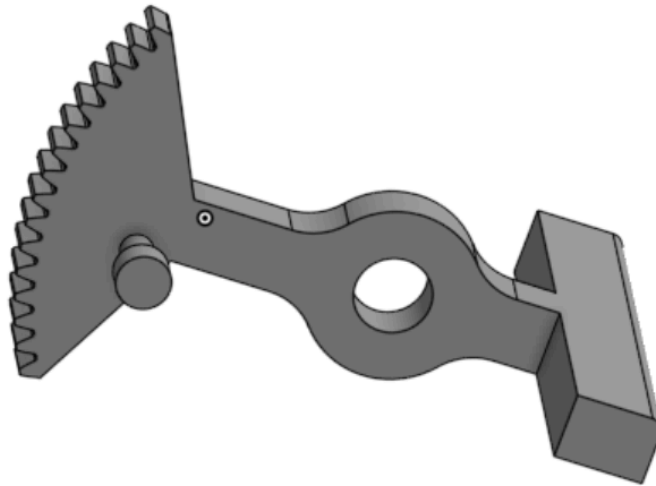


Figure 8: Final iteration of Lever Gear

After determining major flaws within the previous two designs through manufacturing and testing, the final iteration addressed all issues. This design contains a larger hole in the middle in order for a bearing to be fixed inside. The bearing serves as a form of removing most of the friction previously caused by an axle which contacted the plastic lever directly. Furthermore, the spring hole was changed to a peg which allowed ease of attachment and less interference of the spring/rubber band on the lever. Finally, the cam contact arm on the lever was redesigned with greater structural integrity in mind.

Flywheel

The final subsystem of Homie Rektifier served as a means of translating the output rotations of the small output gear of the sliding gear subsystem into constant rotation. This was achieved through a large flywheel which has a large moment of inertia. At the end of the flywheel's axle is a gear which meshes with the small output gear of the sliding gear subsystem, thus enabling the flywheel to rotate and store momentum from the sliding gear subsystem. At the

other end of the axle is a plate which was used as a means of tracking the output speed. This subsystem only needed one iteration.

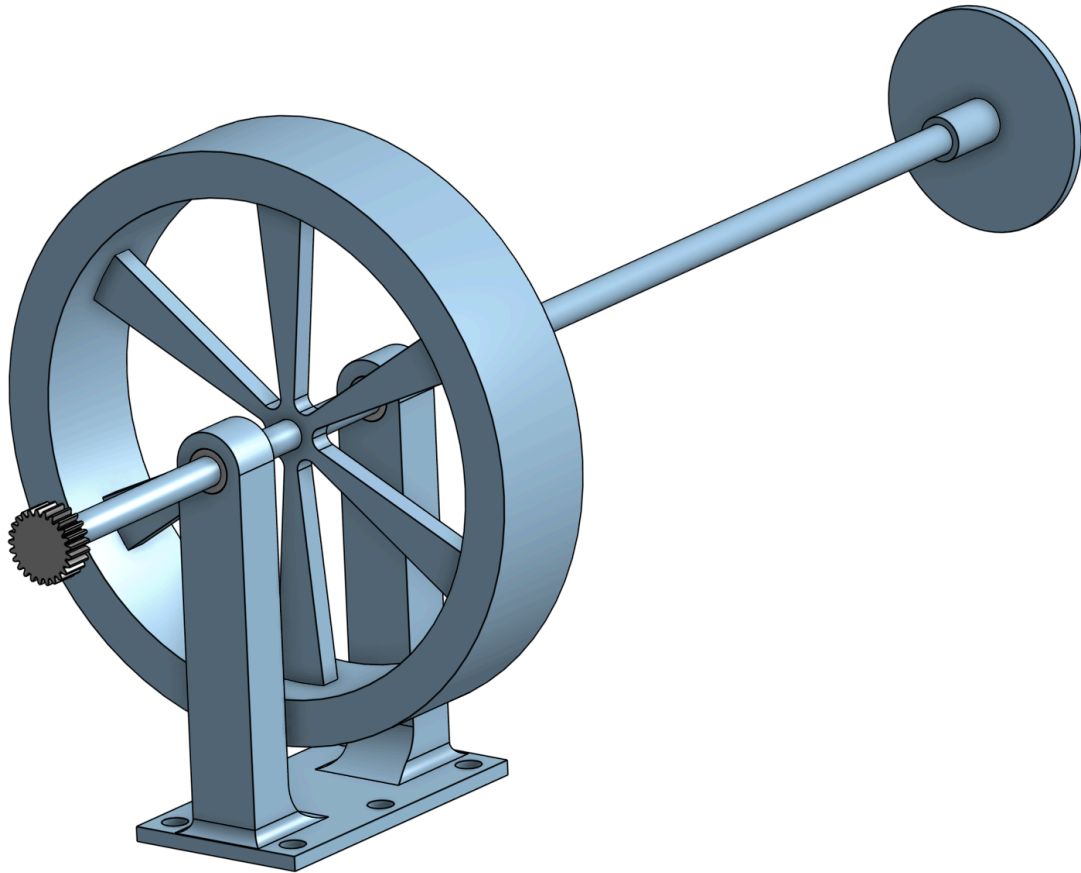


Figure 9: Flywheel subsystem

Homie Rektifier Design Analysis

The Homie Rektifier transforms rotational energy into unidirectional pulses of power which ultimately rotate an output rod. In order to design a system which effectively carries out this rectification, three subsystems were designed, each serving a unique function in the rectification chain.

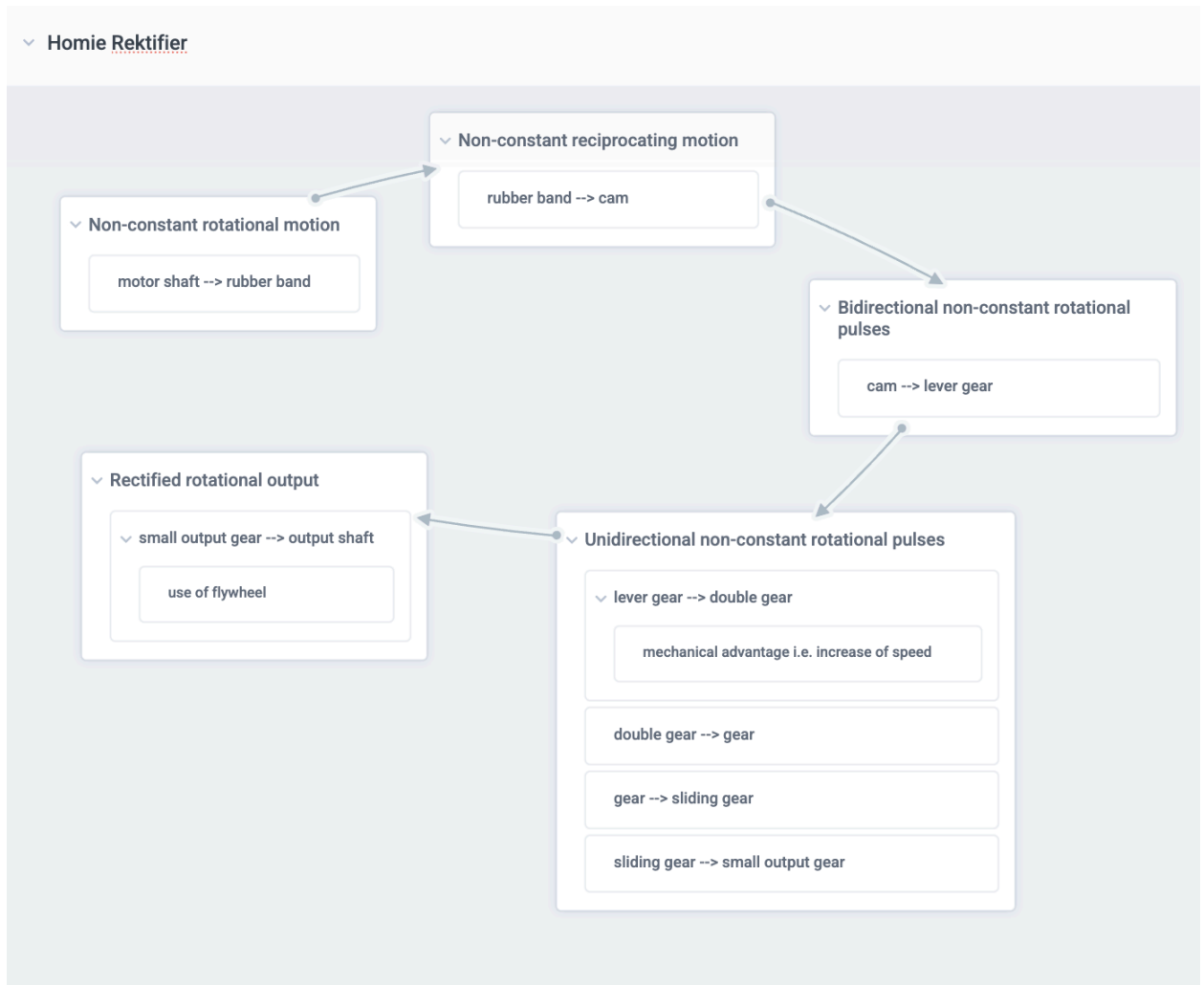


Figure 10: Homie Rektifier Systems Diagram

The first subsystem of Homie Rektifier is the cam to lever gear subsystem. This subsystem begins with rotational power transfer via a rubber band to a more optimal position on the breadboard. The rubber band transfers rotational motion from the motor shaft to the cam. Due to the lack of space on the breadboard, this lateral transformation of power is essential as it allows space to fit the other remaining subsystems. This power transfer also occurs on two wheels of the same diameter, so there is no difference in output speed from the motor shaft to the cam wheel.

The cam wheel is positioned above the breadboard by a vertical support as well as an axle which is supported on the lever gear supports. This configuration implies high stability of the cam and lever arm when power is applied. The elevated position of the cam also allows the lever to pivot comfortably without grazing the breadboard or any other supports.

The lever gear translates the reciprocating motion of the cam back into rotational motion via the teeth on the gear, which is meshed to the next subsystem.

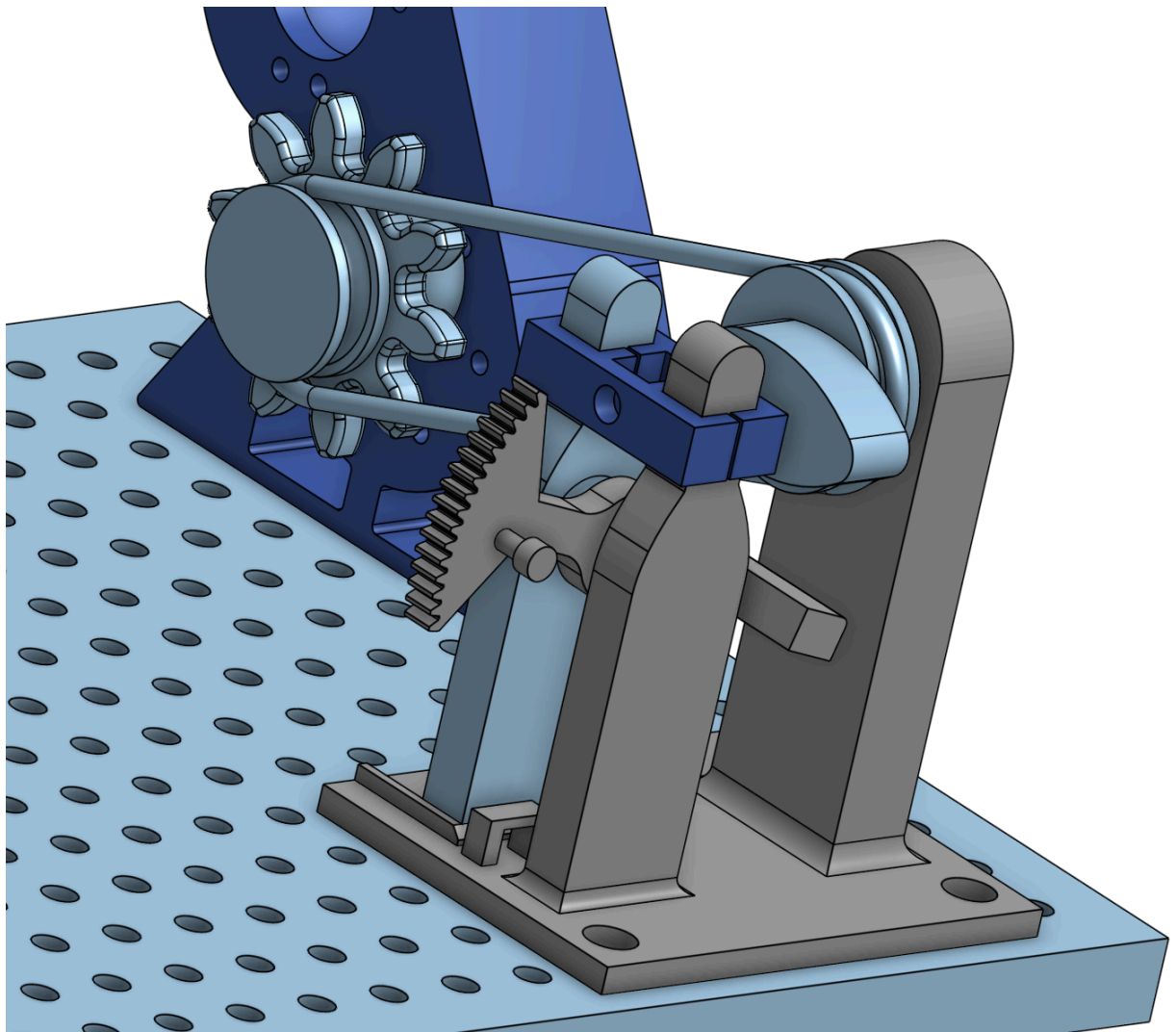


Figure 11: Cam to Lever Gear Subsystem

The following subsystem in the rectification chain is the sliding gear mechanism. This subsection is designed around a method of changing bidirectional rotational input into unidirectional rotational output. The power transfer through the subsystem begins with the large gear. The large gear is physically connected to a gear of smaller size on the other side. This implies that the angular velocity of both gears will be the same; however, because the larger gear has a larger radius, the corresponding linear velocity of the larger gear to another gear will be greater. With this concept in mind, the input of the lever gear onto this smaller gear will return a faster speed of the larger gear. This is important to the design as the sliding gear is only capable of sliding when exposed to high rotational velocity. The following gear in series (dark blue) merely serves as a means of activating the sliding gear. Because of this vertical setup, the sliding gear will naturally position itself disengaged due to the unstable equilibrium of its position. In other words, gravity will force the sliding gear to position itself to the right when at rest. However, when in contact with rotational power, the sliding gear will rotate as well as translate to the left, as there is a free pathway for it to slide. As a result, this engages the final output gear briefly and causes it to spin. Due to the nature of the lever gear fluctuating directions of power repeatedly, this process of the sliding gear engaging and disengaging with the output gear is also repeated. The result of this subsystem is a small gear which receives unidirectional pulses of power.

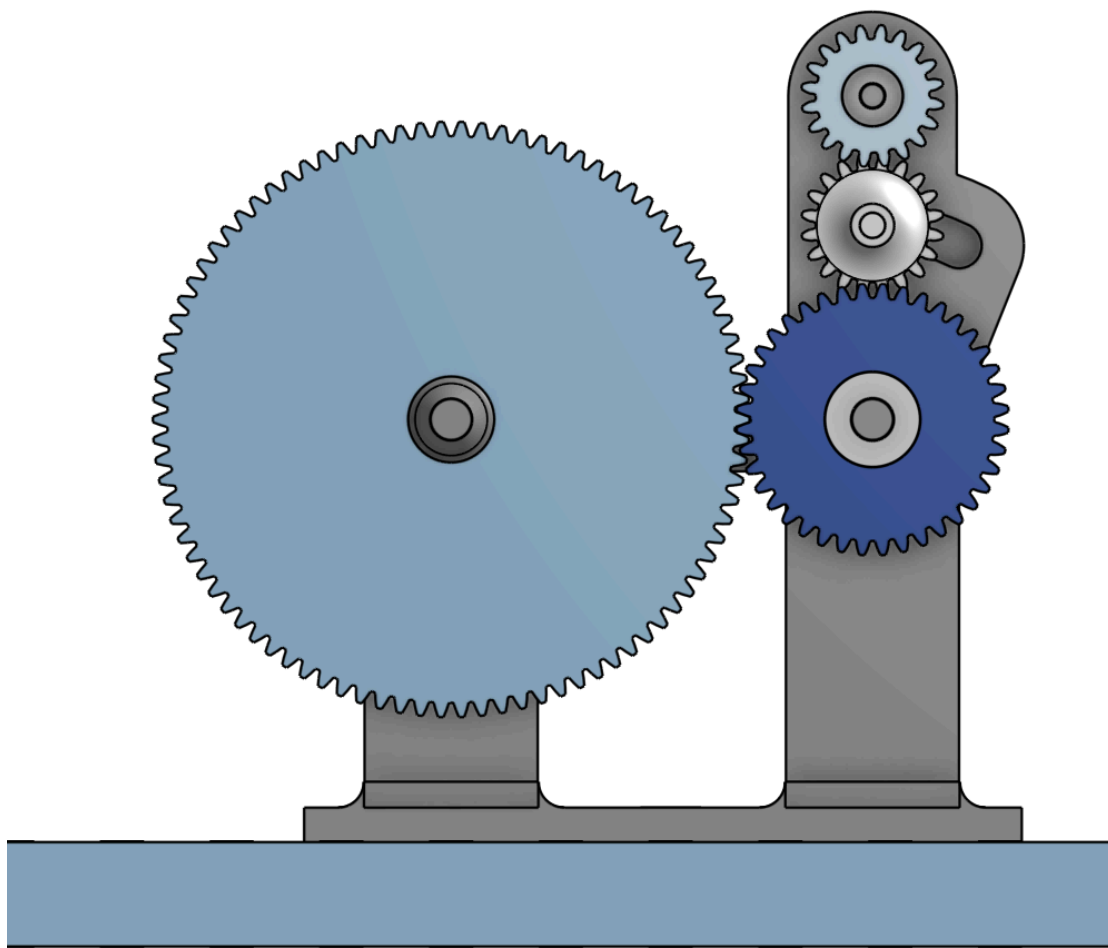


Figure 12: Cross sectional view of sliding gear section

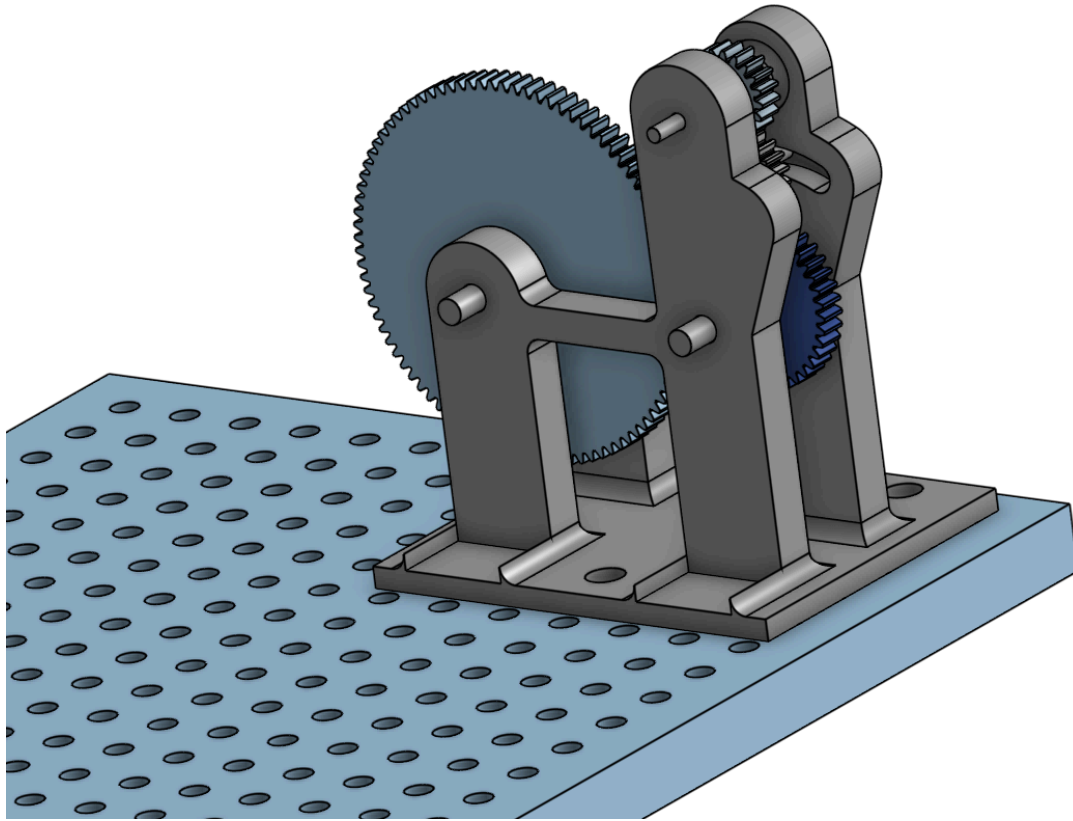


Figure 13: Sliding Gear Subsystem

The final subsystem of Homie Rektifier is the flywheel subsystem. This subsystem receives unidirectional rotating pulses from the small output gear of the sliding gear subsystem and translates it into constant rotational output. This is achieved by attaching an output shaft with a gear on the end of it which engages with the small output gear and thus rotates the shaft. Proper rectification, however, depends on the inclusion of a heavy flywheel with a high moment of inertia. By increasing the moment of inertia through mounting the flywheel, the rod will spin at a steady rotational speed every time it receives a pulse of rotational motion. This occurs regardless of varying subsequent pulses of power. This is similar to the example of the bicycle wheel discussed in the previous section; a high moment of inertia allows rotational motion to be stored

once spun. We used this property to our advantage to ensure smooth mechanical rectification; repeated pulses of power, regardless of frequency, will thus spin the output shaft at a constant speed.

Subsystem	Part Name	Quantity	Description
Encoder Gear	Encoder Pulley Gear	1	Pulley that connects the motor to the lever subsystem
Lever Gear	Lever Gear	1	Lever with gear teeth that engages with the cam to convert motion
	Cam Pulley	1	Rotating cam attached to a pulley
	Thin Rubber Band	1	Used in tension to return the lever to its original position
	Thick Rubber Band	1	Used to transmit rotation from encoder gear to cam
	Lever Stand	1	Provides an axis of rotation for the cam and the lever gear
	Lever Support Beam	1	Prevents the lever gear from sliding off
	Lever Support Beam Bracket	2	Holds the support beams between the lever gear together
	0.25" Screws	3	Secures the stand to the breadboard
	3ID 7OD 3OAL Bearing	1	Allows the lever to spin around the rod freely
	5ID 11OD 5OAL Bearing	1	Allows the cam to spin around the rod freely
Sliding Gear System	Sliding Gear	1	Slides back and forth to connect motion from input gears with output gears
	Spur Gear - 90 teeth	1	First input gear driven by the lever gear to transmit motion
	Spur Gear - 40 teeth	1	Second input gear driven by the first input gear
	Spur Gear - 20 teeth	1	Output gear driven by the sliding gear
	Gear Stand	1	Vertical stand with rods for the gears to spin on
	Removable Gear Support	1	Interlocking vertical support that keeps the gears in place
	Gear Support Beam Bracket	1	Holds the gear stand and the gear support together
	5ID 11OD 5OAL Bearing	2	Allows the input gears to spin around the rods freely
	3ID 5OD 5OAL Bearing	1	Allows the output gear to spin around the rod freely
	0.25" Screws	3	Secures the stand to the breadboard
Flywheel	Flywheel	1	Heavy wheel mounted to output shaft that stores rotational energy
	Spur Gear - 25 teeth	1	Mounted to the output shaft; driven by the final output gear to rotate the output shaft
	Flywheel Stand	1	Vertical stand to support the outshaft and flywheel
	8ID 12OD 3.5OAL Bearing	2	Allows the output shaft to spin freely
Output Shaft	0.25" Screws	4	Secures the stand to the breadboard
	8mm Metal Shaft	1	Metal rod used as the output shaft

Figure 14: Bill of Materials

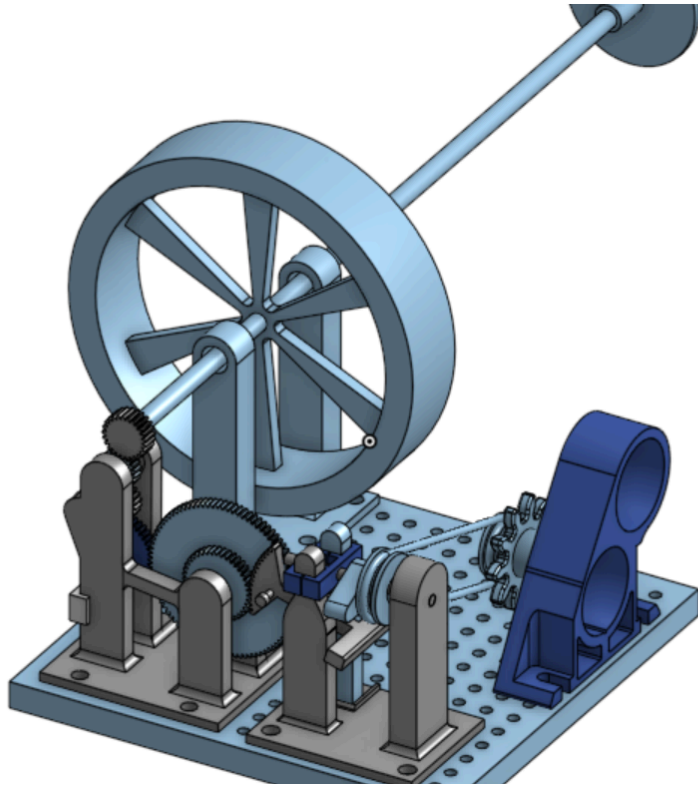


Figure 15 : Homie Rektifier Full Assembly

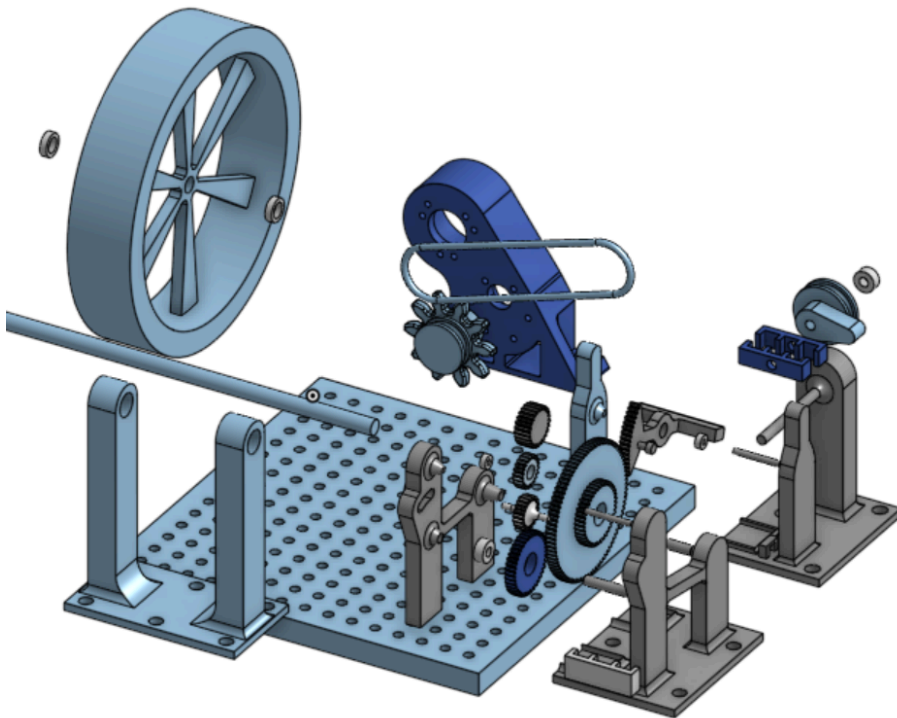


Figure 16 : Homie Rektifier Assembly Exploded View

Outcome Analysis

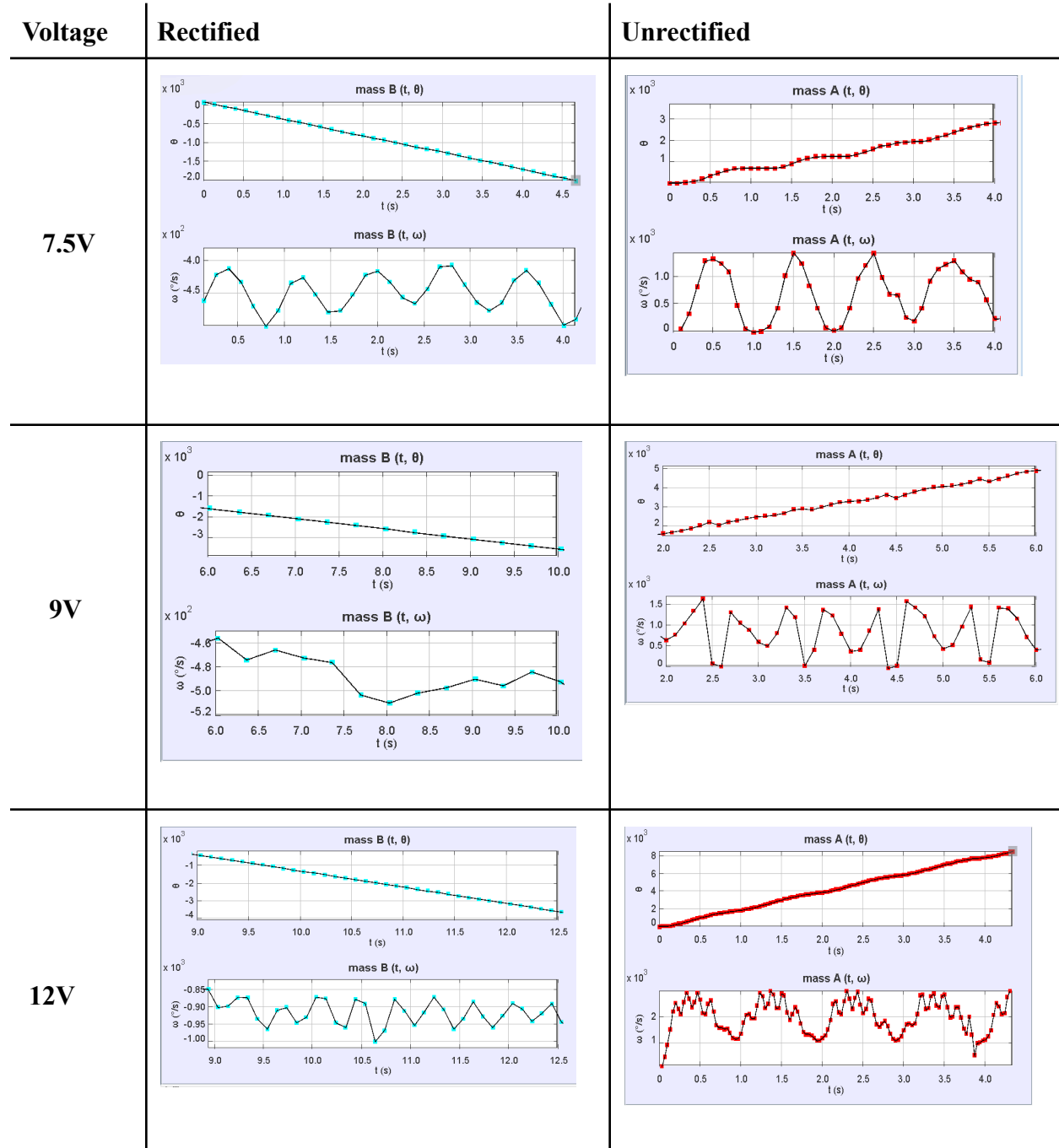


Figure 17: Rectified Vs Unrectified Angular Velocity

Based on the graphs in figure 17, the Homie Rektifier system proved to effectively rectify across every tested voltage level. When comparing the rectified and unrectified graphs outputs at

7.5V, 9V, and 12V, the rectified system consistently demonstrated smoother, more stable rotational motion.

When analyzing the angular displacement (θ) of the rectified system, the graph displays a clean, linear trend with little to no visible bumps. Meanwhile, the unrectified plots have large visible bumps and overall exhibited step-like behavior, suggesting varying torque and motion. The most glaring evidence of Homie Rektifier's effectiveness is the angular velocity graphs (ω). The rectified outputs showed significantly reduced fluctuation and more uniform wavelengths, indicating a more constant rotational energy transfer. In contrast, the graphs of the unrectified system produced erratic angular velocity.

By also analyzing the amplitudes of the graphs, the unrectified graph exhibited a much greater amplitude of angular velocity than the rectified graphs at every voltage level. At 9 volts, for instance, the amplitude of the rectified and unrectified graphs ranges from 480 degrees/sec to 510 degrees/sec and 0 degrees/sec to 1,420 degrees/sec, respectively. This pattern was consistent across all voltage levels. A rough summary of the observed angular velocity ranges and stability is shown below:

Voltage	Rectified ω range ($^{\circ}/s$)	Unrectified ω range ($^{\circ}/s$)
7.5 V	420-480	0-1150
9 V	480-510	0-1420
12 V	850-970	750-2500

Table 1

The table clearly illustrates that the rectified system produced a significantly more stable output across the board. The sharp contrast in amplitude range between the rectified and unrectified systems demonstrates how effectively the mechanical design filtered the input fluctuation.

In addition to amplitude control, the rectified system also reduced the effective frequency of output fluctuations. While the unrectified angular velocity plots displayed high-frequency spikes that increased with input voltage, the rectified plots remained periodic and low in frequency, indicating consistent filtering. This suggests the rectifier dampens both the intensity and rate of fluctuation, contributing to a much smoother output motion.

Overall, the Homie Rektifier successfully met its design objective of transforming non-uniform input motion into a nearly constant output speed. The evidence from both graph analysis and quantitative comparisons supports this result, and Homie Rektifier's consistent performance across multiple voltages demonstrates its validity and reliability.

Future Improvements & Conclusion

There are a few changes which could be made to make this system more effective, reliable, and aesthetically pleasing. Firstly, although the pulley system used to transfer power from the motor gear to the cam gear is technically functional, it undergoes a lot of slipping and is undoubtedly not the best solution. It may be more logical to employ an intermediary gear in its place.

The tolerances on almost every axle throughout the rectifier could be improved on, as most were not ideally fitted to the bearing or part they were meant to interact with. For example, the flywheel had to be held onto its axle with tape, which was a suboptimal and rushed solution. An effective way to fix this issue would be to print several rods and rings of slightly varying diameters all at once and test these samples with the other parts to determine the necessary tolerances. On top of this, further experimentation could also be done with different gear ratios and flywheel designs to potentially discover ways to increase the device's efficacy even more.

Aesthetically, no attention was given to the rectifier's color, and some prints had to be done with different colored PLA due to time pressure. In the future, a color scheme could be selected to make the machine more pleasant to look at.

To conclude, the Homie Rektifier has proven functional at its task of converting non-constant unidirectional rotary motion to constant velocity unidirectional rotary motion. At 9V, the most effective voltage for the rectifier, the rectified range in angular velocity was 30 °/s (degrees per second), compared to an unrectified range of 1420 °/s. This was a very drastic reduction in oscillation, indicating the machine functioned as desired. At 12V, the least effective voltage, the rectified and unrectified ranges in velocity were 150 °/s and 2000 °/s, respectively. The rectifier was assembled well within five minutes and able to start up within 10 seconds, meeting all of the project requirements.

Many iterations were made throughout the design process, and several failures were necessary in order to reach success. For example, the first cam and follower printed didn't line up at all and an almost entirely new design had to be drawn up in CAD, and roadblocks like this were typical as the rectifier was being built. The team learned to troubleshoot effectively, create well-organized CAD parts and assemblies, optimize 3D print quality and tolerances, and work with various forms of power transfer. In addition, the project taught us systems thinking skills and how to take a creative, logical approach a constrained problem.